MILITARY SPECIFICATION

PROJECTILE, 155MM: M825 METAL PARTS ASSEMBLY

This amendment forms a part of Military Specification MIL-P-64008 (AR), dated 27 December 1982, and is approved for use by the US Army Armament Research and Development Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Change the title to "Projectile, 155MM: M825 Metal Parts Assembly."

1: Delete "cope" and substitute "Scope".

1.1 Delete "XM825" and substitute "M825".

PAGE 2

Under Product and Packaging drawing 92352634 delete title and substitute "Projectile, 155MM, M825, Metal Parts".

Under Inspection Equipment drawings make the following changes:

"T9280354 - Eddy Current Standard, XM825 Base" to:

"9280354 - Eddy Current Standard M825 Base" and

"T9280355 - Envelope drawing Eddy Current Equipment, XM825 Base" to:

"9280355 - Envelope drawing Eddy Current Equipment, M825 Base".

* 2.2: Under "American Society for Testing and Material (ASTM)" Add the following:

"ASTM Method E-18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials"

FSC 1320

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PAGE 4

* 3.4.1: Delete and substitute the following:

"3.4.1 Microstructure. The microstructure of the ogive and base shall comply with the microstructure depicted in Figures 1A and 1B or 2. Two examples of acceptable microstructure are shown in Figures 1A and 1B. In Figure 1A, the insoluble phase is well broken up, randomly distributed within a matrix which is essentially unrecrystallized. In Figure 1B the insoluble phase is well broken up and randomly ditributed within a matrix which is partially recrystallized. An example of a marginally acceptable microstructure is shown in Figure 2. In this figure the insoluble second phase exhibits some discontinuity and appears to be elongated. An example of an unacceptable microstructure is shown in Figure 3. The insoluble phase is distributed in a more continuous form (Network) within a matrix which shows a large degree of microsegration (or coring). The microstructure in Figure 3 must be avoided because it is detrimental to the integrity of the material."

* Add the following paragraph:

"3.4.2.1 Base heat treatment verification. Each base shall be Rockwell B85 minimum."

PAGE 5

3.7: Delete "XM825" and substitute "M825".

* 3.8.1: Second line, insert "microstructure," between "process" and "and".

PAGE 11

4.4.2.a: Add the following to the end of the last sentence:

"or MIL-STD-1235, Inspection Level II where applicable".

PAGE 19

4.4.2.4:

Major 131: Delete in its entirety.

Major 135: Delete "Gage" and substitute "Visual".

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PAGE 20

4.4.2.4:

Major 147: Under Examination or Test, delete "&".

Add the following major examination:

"148. Functional runout of body cavity 0.40% 3.2 Gage"

PAGE 24

4.4.2.6:

Major 119: Delete in its entirety.

Major 121: Delete "Gage" and substitute "Visual".

PAGE 25

4.4.2.6, add the following major examination:

"127. Taper incorrect 0.40% 3.1 Gage"

PAGE 26

* 4.4.2.7, delete major category 103 in its entirety.

PAGE 28

4.4.2.7:

Minor 201, under Examination or Test, add "diameter" after basic".

Minor 203, under Examination or Test, delete "Diameter on taper" and substitute "Taper incorrect".

Minors 204 and 205, delete "Gage" and substitute "Visual".

PAGE 29

4.4.2.8, Major 104: Delete "Gage" and substitute "Visual".

4.4.2.8: Delete Majors 104, 106 and 107 in their entirety.

PAGE 31

* 4.4.2.9: Add the following major categories:

"107 Lifting plug or gasket missing - 100% 3.1 Visual
108 Lift plug detorque - 0.40% 3.1 Gage"

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PAGE 32

* 4.4.2.9: Minor category

208: Delete "& fuze"

PAGE 33

* 4.4.2.10: Major category delete "None defined" and add the following:

"101 Support cover missing (8 places) - 100% 3.1 Visual"

PAGE 34

* 4.4.3.3.1: Delete and substitute the following:

"4.4.3.3.1: Base and ogive. First article bases and ogives shall withstand the salt spray test as prescribed in para. 3.10.1.2 of MIL-A-8625. The production lot bases and ogives shall comply with para. 3.10.1.2 of MIL-A-8625 except withstand 72 hours and show no evidence of corrosion spots and/or pits on any surface except within 1/16" of surface edge. Corrosion on electrical contact points shall not be cause for rejection."

4.4.3.3.3: Delete "XM825" and substitute "M825".

PAGE 36

* 4.5.1.3: Delete and substitute the following:

"4.5.1.3 Verification of heat treat of base. The test shall be made using procedures as prescribed in ASTM E-18. The readings shall be taken on the surface indicated on the applicable drawing."

PAGE 38

4.5.3.2: Add the following after the first sentence:

"At the option of the contractor, this test may be performed prior to machining of the rear counterbore, keyway/threads".

PAGE 39

* 4.5.5: Delete the next to the last sentence.

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PAGE 40

* 4.5.7: Delete and substitute the following:

"4.5.7 Microstructure examination (ogive & base). One ogive & one base forging from each heat treatment lot shall be selected for microstructure examination. The metallographic specimen shall be etched with Keller's reagent and all areas examined under a microscope/metallograph at a minimum magnification of 250 diameters. Any evidence of the insoluble phase generally occurring in a semi-continuous network form shall be cause for rejection of the lot represented by the sample. Microstructural preparation and examination shall be performed by a metallographer or metallurgical engineer familiar with aluminum alloys."

PAGE 44

* Delete title, "Figure 2 Unacceptable Microstructures (250X)"

* Delete "a" and substitute "Figure 2. Marginally Acceptable Microstructure (250X)"

* Delete "b" and substitute "Figure 3. Unacceptable Microstructure (250X)"

PAGE 45

* 5.1.1, 5.1.2, and 5.1.3, Delete in their entirety.

PAGE 46

* 5.2 Delete and substitute the following:

"5.2 Packaging, packing and marking."

* 5.2.2: Delete and substitute the following:

"5.2.2 Packaging, Packing, and Marking of Bases shall comply with the following:"

* Add the following paragraphs after paragraph 5.2.2.

"5.2.2.1 Overseas shipments. Bases shall be semi-bulk or single item packaged, moisture proof packaged (barrier bag only), packed with cavity facing downward, marked, and palletized in accordance with MIL-STD-1169, Level A. Bases shall be separated or partitioned (if bulk) in the unit package to protect the paint surfaces and shall include marking to assure that the base is shipped or stored with cavity facing downward."

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"5.2.2.2 Domestic shipments. Bases shall be semi-bulk or single item packaged, packed with cavity facing downward, and marked in accordance with MIL-STD-1169, Level B. Bases shall be separated or partitioned (if bulk) in the unit package to protect the paint surfaces and shall include marking to assure that the base is shipped or stored with the cavity facing downward. Palletization is optional."

PAGE 47

6.6: Delete "XM825" and substitute "M825".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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